Rasmus Duret

8 August 2024

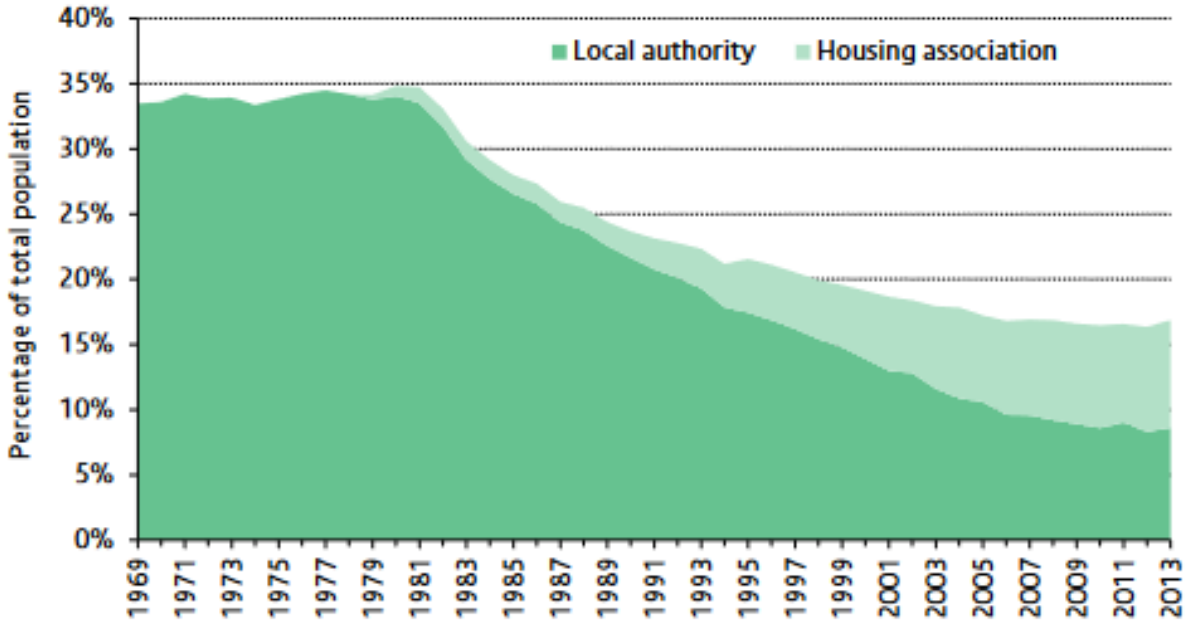
@TheIFS

Summer Student Project: Effects of Social Housing

Economic and Social Research Council

Motivation

Figure 1. Percentage of the population of Great Britain living in social housing



Source:
IFS, 2015.

2024 Labour Manifesto pledges to *"deliver the biggest increase in social and affordable housebuilding in a generation."*

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Overview

Question: **What are the effects of moving into social housing?**

Data: **Understanding Society**, waves 1-5.

Methodology: **Event Study**, comparing movers with future movers.

Findings: **Overall, positive effects**, albeit imprecisely estimated.

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Literature Overview

Outcomes

- Within economics, mostly concentrated on material outcomes: schooling results, earnings, benefit receipt, employment…
- Elsewhere, work on other outcomes: physical and mental health, wellbeing, social networks…

Methods

- Most convincing papers use large-scale experiments (US, Moving to Opportunity) or quasi-experimental systems (Amsterdam lottery).
- Using quasi-random variation in allocation rules or timing is common, but **often challenging** (limited data, weak instruments).

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Literature Overview

Moving into SH generally found to have negative effects.

- However, often found to be driven by "bad neighbourhoods".
- Moving to SH in "good" areas can have positive effects (see Van Dijk, 2018).

UK context not extensively studied.

- But concerns a substantial population (~17%, 12M).
- How are these individuals affected by the modern SH system?
- Large variation in systems, contexts… → **Limited external validity**.

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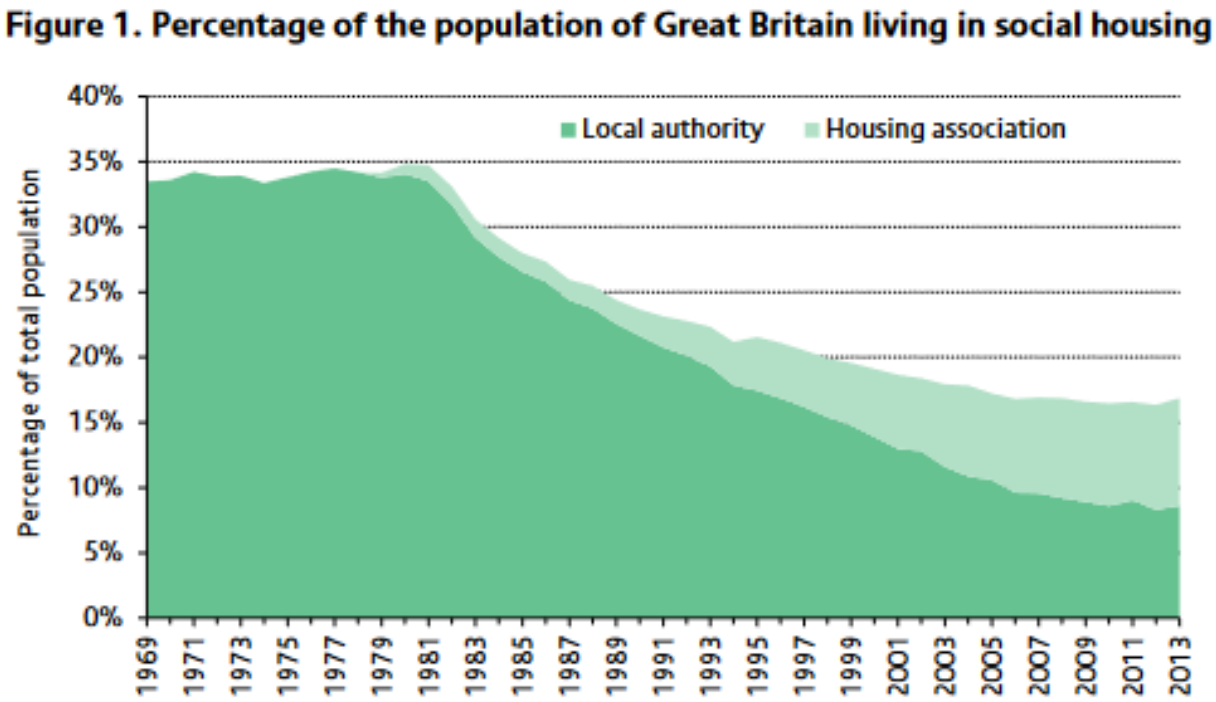
Background

Large fall in SH since 1980



Source: IFS, 2015.

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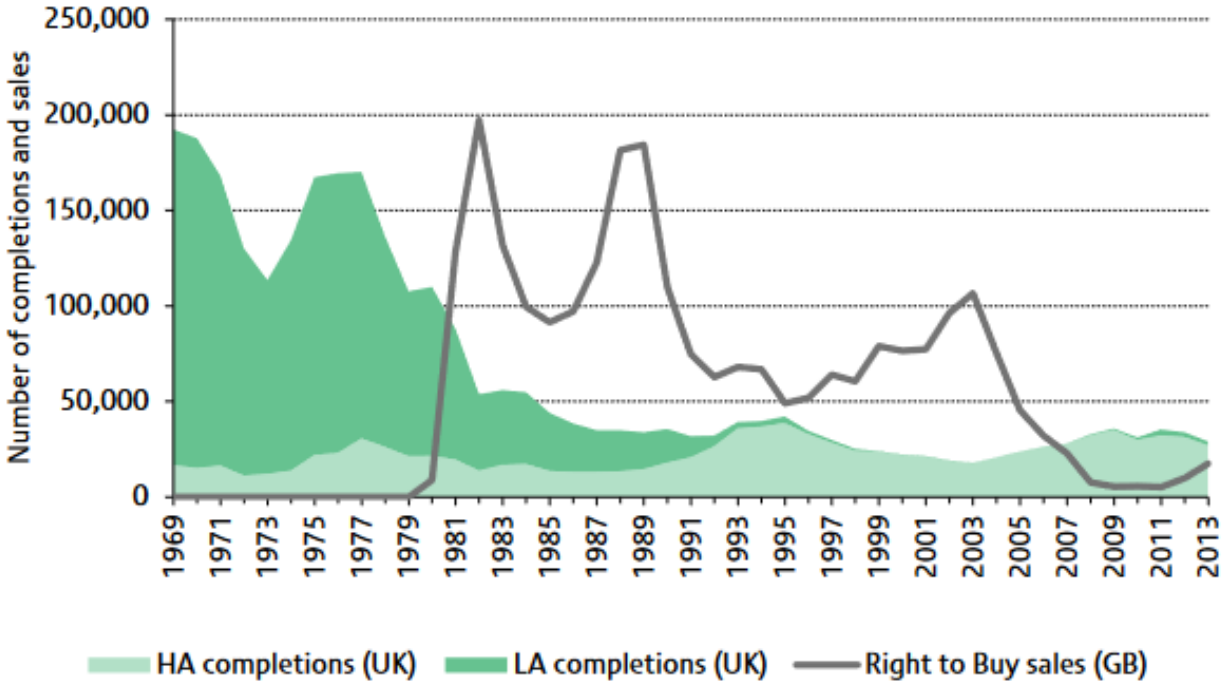
Large fall in SH since 1980

Figure 2. Social housing completions and Right to Buy sales since 1969



Source: IFS, 2015.

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Broad Periods

IFS

1980s: Privatisation and Deregulation

- Transfer to Housing Associations and private owners (Right to Buy).
- Cost cutting for public entities, private landlords strengthened.

1990s/2000s: More duties and quality control

- New housing duties: **LA's must provide for those in need**.
- Controls on SH quality and price.

2010s: Decentralisation, narrower targeting

- Delegation of duties/powers to LA's: allocation rules, prioritisation.
- *Subsidise individuals when they need it most*.

SH has become more targeted towards individuals in need, with more duties but also more flexibility given to individual LA's.

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The system today

Local Authorities have a duty to provide housing for those “in need”

→ Mental/physical health; homelessness; violence; overcrowding…

Local Authorities create their own allocation and selection criteria

- Not intended to target those not in specific need.
- Often points- or rules-based.

→ Potential for quasi-random variation in allocation if good data.

This is the institutional context in which I conduct the analysis.

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Data

Understanding Society

Objective: Study moves into SH

→ Need to track people over time, their tenure, and outcomes.

Understanding Society

- Panel survey, 13 waves starting in 2008.
- Outcomes: Net/gross rent and income; hours worked, employment, mental health proxy (GHQ scores).
- Main sample: Individuals aged 25-60 in 2008.

Define: Social Housing = Local Authority + Housing Association.

Cannot identify moves within SH.

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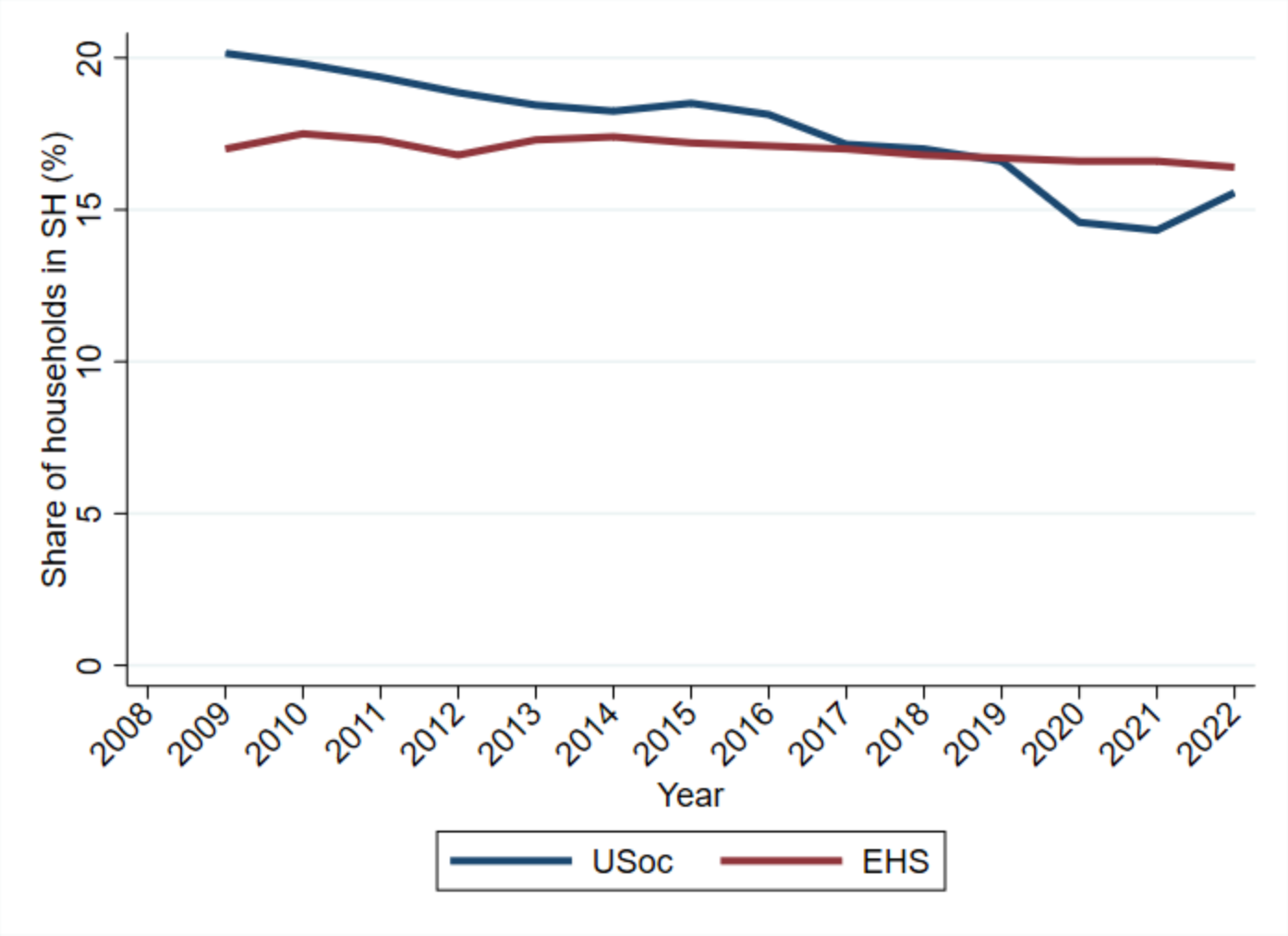
Matches national share in SH



Share of individuals with fully observed tenancies. SH includes housing associations and local authority rents.

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Methodology

Social Housing as a "treatment"

"Living in SH" is a package of different treatments

- Neighbourhood: Opportunities? Social capital? Local public goods?
- Subsidised Rent: Income vs Substitution effects?
- Housing quality: Negative effect on health and development?
- Effect of moving: Disruption of social networks?
- Why move: Fleeing unsuitable housing? Seeking opportunities?

→ Difficult to disentangle all these.

Here, very reduced-form:

What are the short-term effects of moving into social housing?

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Empirical Design

Defining Treatment

- Treatment assignment is non-random: based on specific criteria/needs.
- Systematic differences between those likely to be in SH and those who never are.

→ Cannot do a simple comparison or use cross-sections.

Event study: what happens when people move?

- Can test pre-trends, this is relevant to policy evaluation / design.
- Can obtain causal estimates if move-in timing is random.
- Treatment: Movers into SH.
- Control: Future movers, more likely to be comparable.

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Estimation

$$Y_{it} = \beta X_{it} + \sum_{k=-(T-1)}^{T-2} \delta_k + \eta_t + \varepsilon_{it}$$

- Time FE in main spec *(TWFE as robustness check)*.
- Impose $\delta_{-1} \equiv 0$: effects measured by δ_k are relative to pre-move.
- Balanced sample of Waves 1-5, cross-sectional weights from W5.
- $N \approx 400$, depending on outcome.

- Outcomes: Rent, Income, Employment and work hours, GHQ score.
- Controls: Initial values of income, marriage status, # children.

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Causality

Can argue causality if move-in timing is random.

Why might move-in timing **not** be random?

- Response to a shock: unemployment or health.
- Selection on gains: May only choose to move if beneficial.
- Macroeconomic conditions: Likely to affect private market (2008).

→ Overall, not strongly believable, likely to capture some bias.

However, this is also how the system is supposed to work today: SH is supposed to support individuals in times of need, as a *response*.

→ Existence of pre-trends is of policy relevance.

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Descriptive statistics

	Never SH	In SH		
		All	Waves 1-5 (Treatment)	Waves 6-9 (Control)
Monthly Gross Rent	994.68	916.07	921.55	906.92
Monthly Net Rent	927.29	745.25	743.55	748.15
Accom. in decent condition	0.87	0.71	0.71	0.70
# Rooms	5.08	3.95	3.92	4.02
People / Bedroom	1.06	1.31	1.32	1.30
Age	42.30	39.69	40.06	38.99
Married	0.62	0.45	0.49	0.38
# Children	0.55	0.74	0.70	0.82
# Young	0.32	0.32	0.30	0.37
# Adults	2.42	2.40	2.36	2.47
GHQ Score / 12	1.70	2.68	2.54	2.96
GHQ Score ≥ 3	0.21	0.27	0.24	0.32
Monthly Household Income	5955.32	3768.21	3858.16	3598.60
Monthly Individual Income	2917.87	1951.27	1976.29	1903.08
In paid employment	0.80	0.62	0.63	0.59
Weekly work hours	34.28	33.64	34.01	32.86
Observations	23331	1036	677	359

Data based on W1 only. Higher GHQ score indicates worse mental health. All variables not observed for all individuals.

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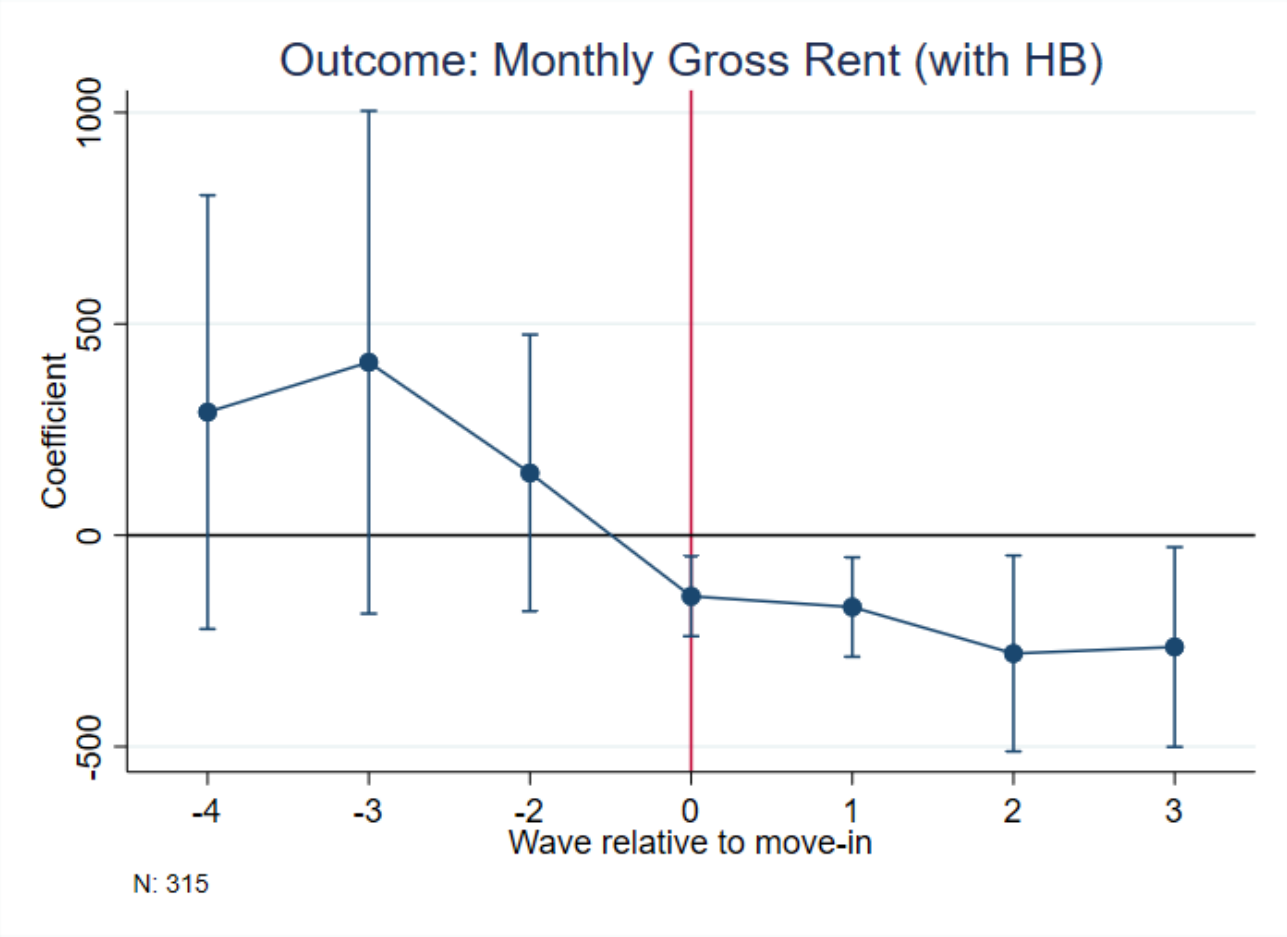
Results

Gross Rent



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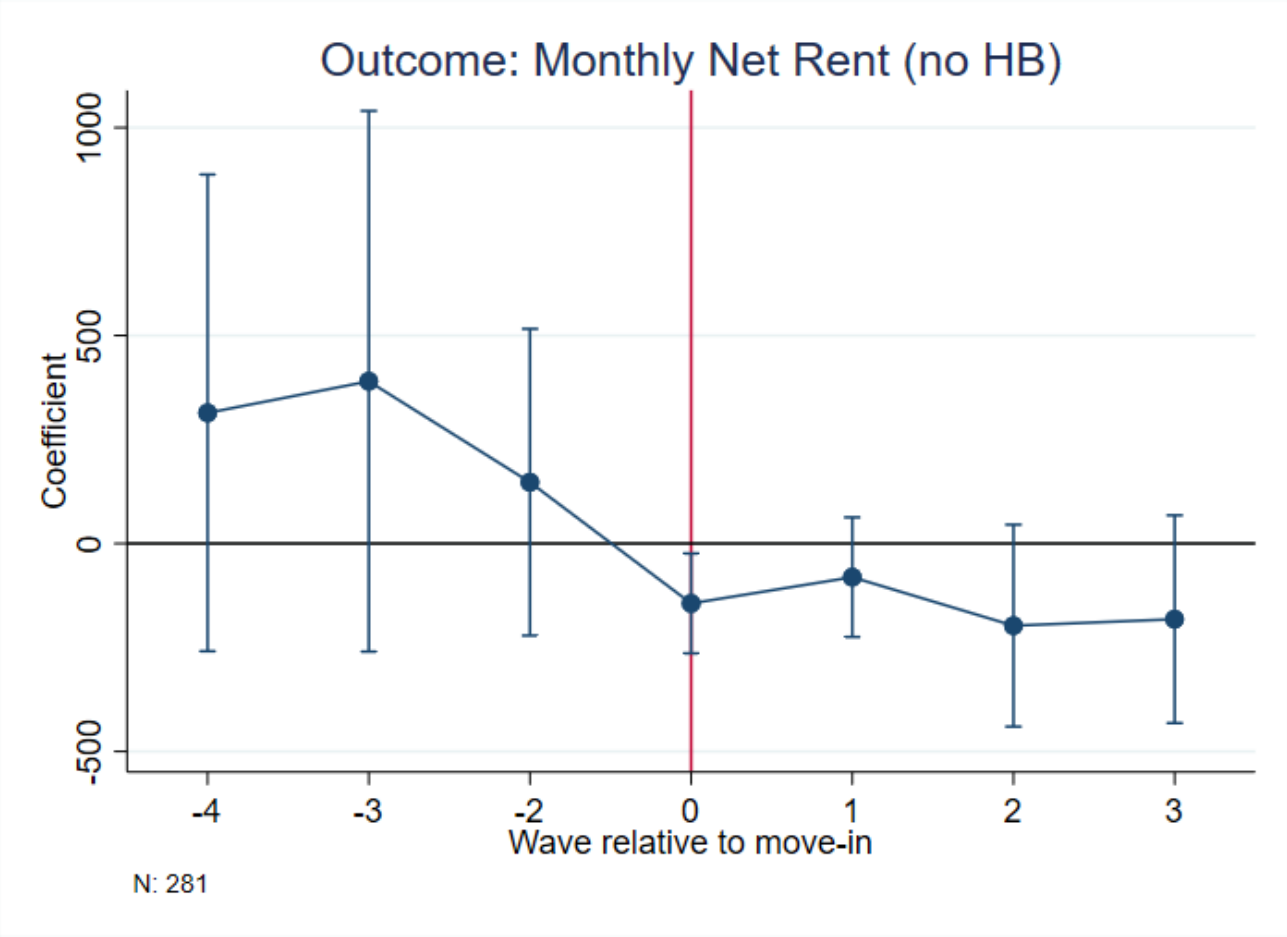
Net Rent



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Monthly Individual Income

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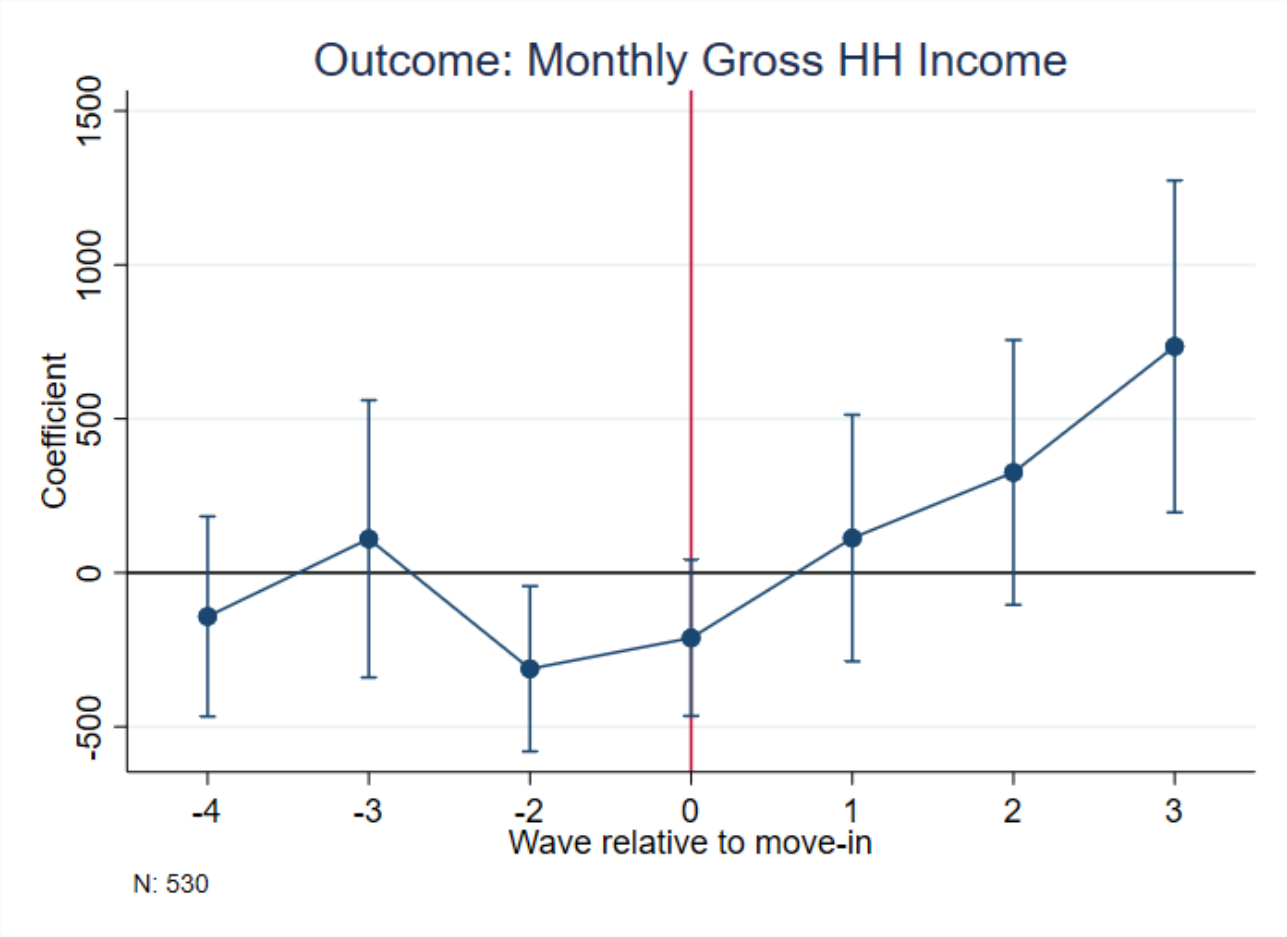
Monthly Household Income



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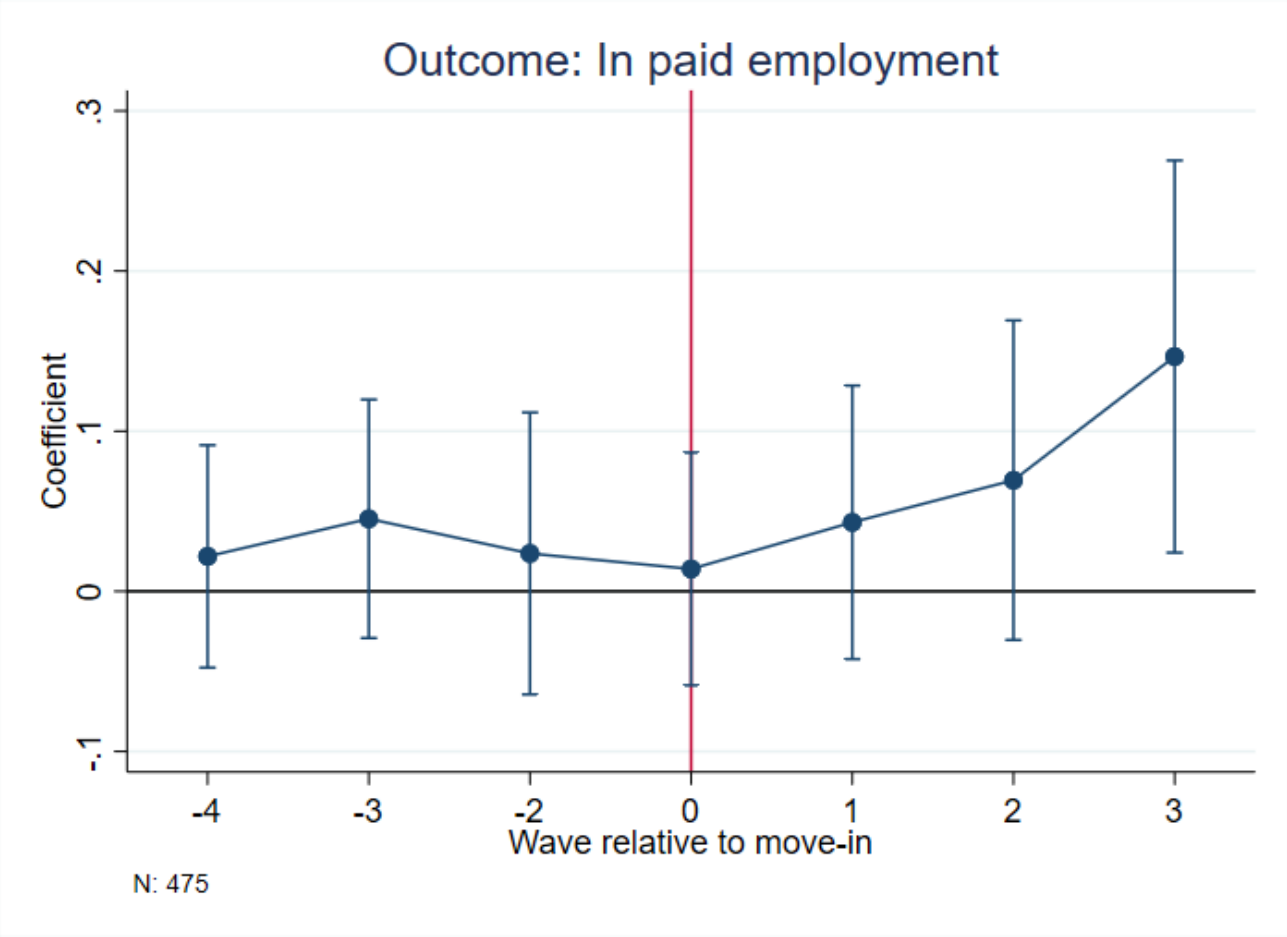
Employment



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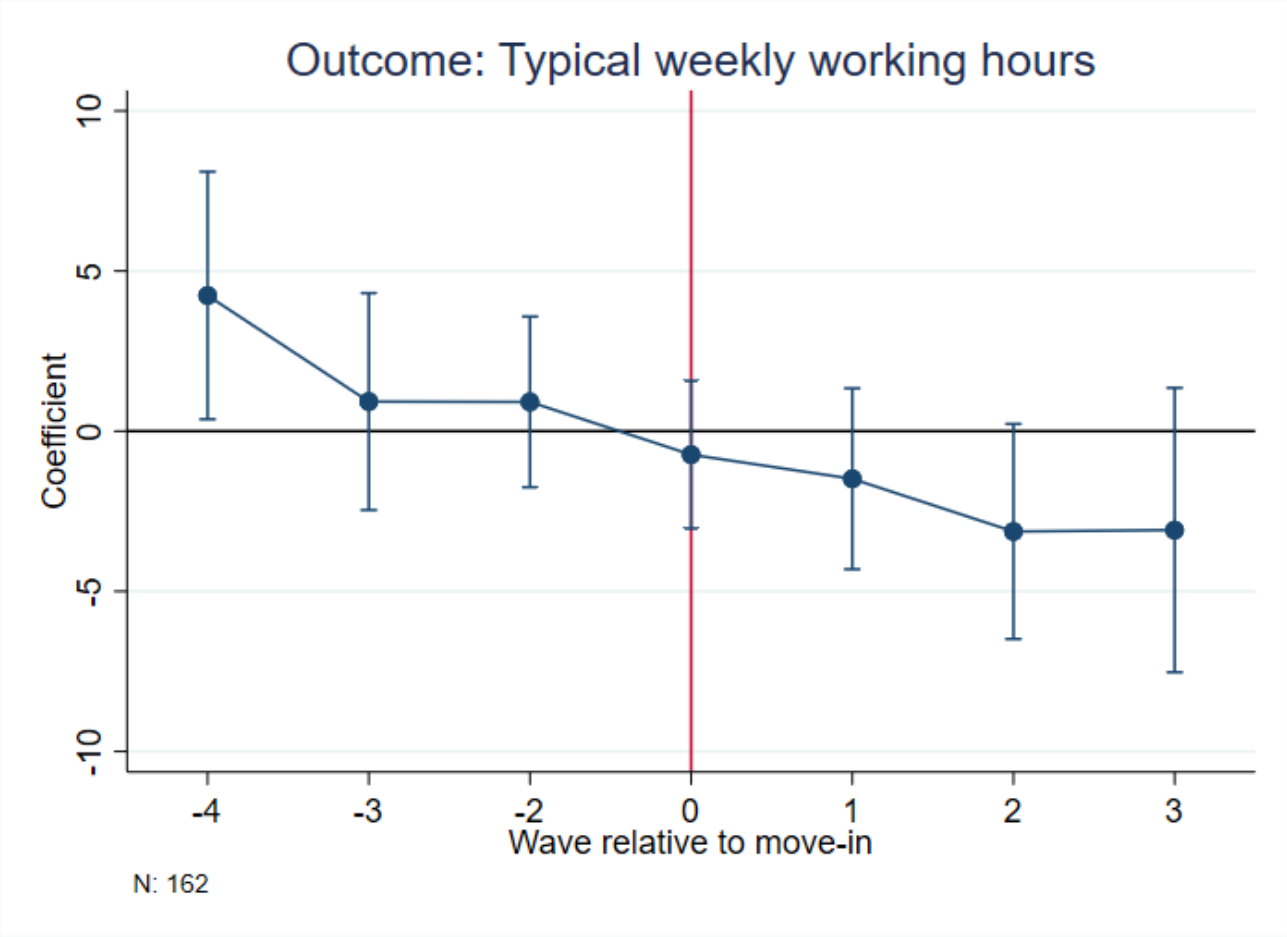
Hours Worked



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P(GHQ Score ≥ 3)

Higher score = higher estimated chance of mental distress. A score of 3 is a commonly used threshold level.

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Discussion

Highly noisy estimates, at best weak pre-trends, small sample.

- Pre-trends are not necessarily a problem if we take moving into SH as a **response** to what happens in the pre-periods.
- In spite of the imprecision, some patterns do appear to emerge.

Overall, it appears that **moving into SH is correlated with short-term improvements** to this set of outcomes, if anything.

- Lower rent in both net and gross terms.
- Higher incomes and employment.
- Improved mental health.
- Pre-trends indicate that SH is indeed some sort of response.

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Next steps

Some additional avenues with more limited data in USoc

- Reasons for moving?
- How far do people move?
- Do people like/feel safe in their new neighbourhood?

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Appendix

More descriptives

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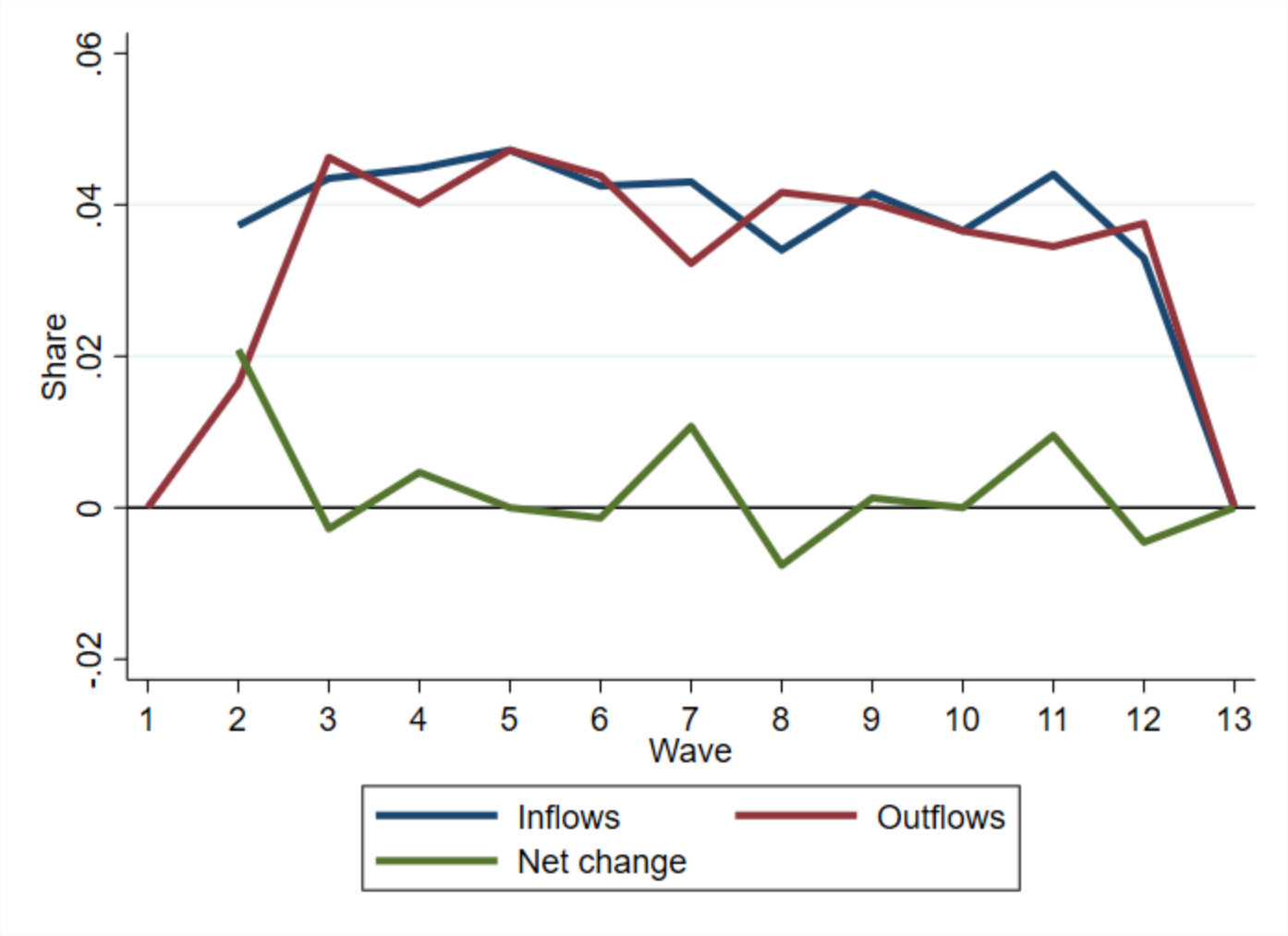
Flows are relatively stable



As share of fully observed tenancies.

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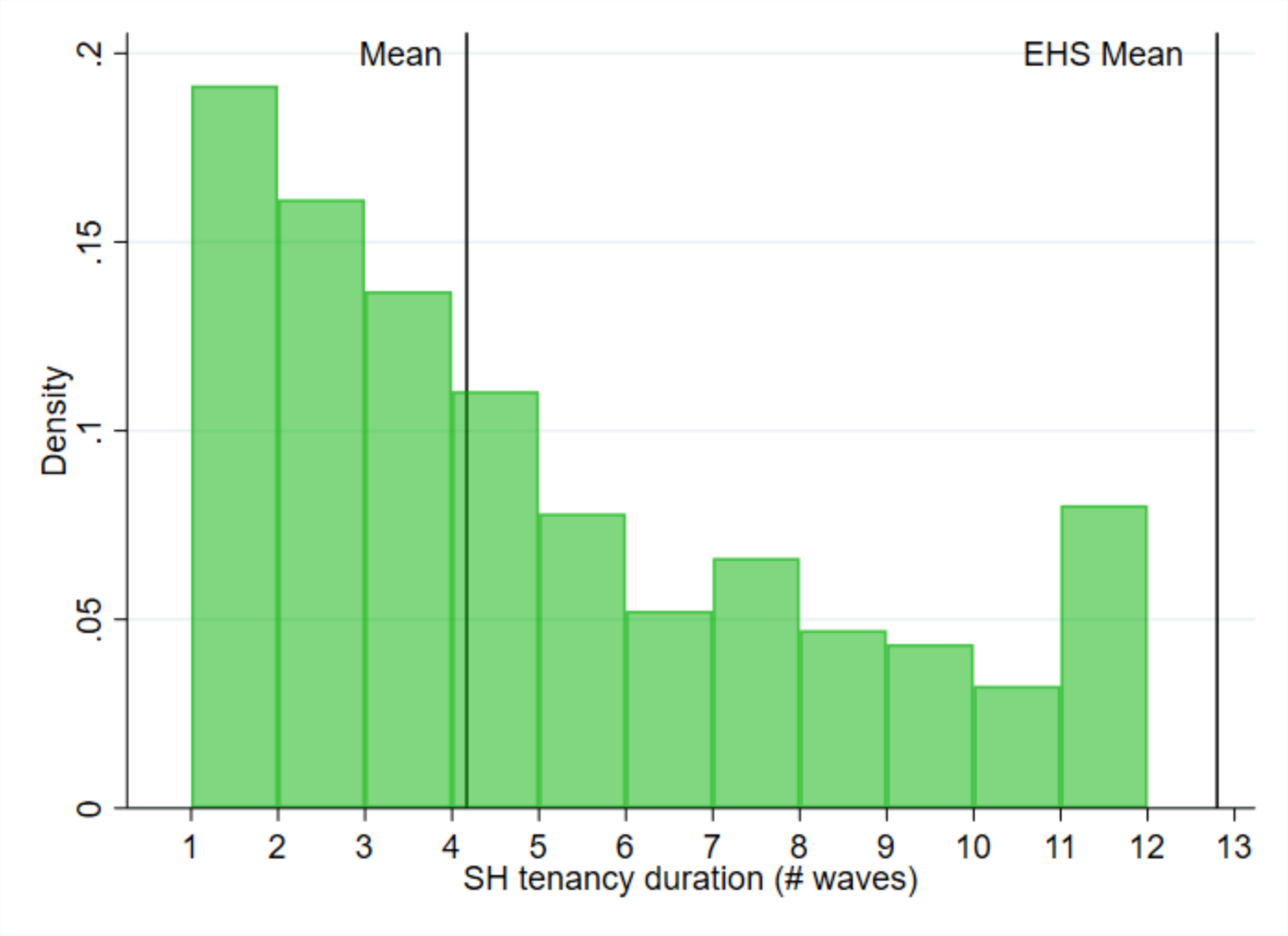
Poor match for spell durations



Fully observed spells only.

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Tenancy Durations

Several reasons why this is a poor match.

- Mechanical: only observe people for 13 years.
- Missing data: Dropping all individuals with any missing observations means we are likely losing a lot of longer spells.
- Sample construction: People in SH in wave 1 are likely to be in SH in other waves, but we drop these.

However, since looking at *short-term* effects for *new* movers, this is not a significant limitation to *internal* validity.

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Sample Construction

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Sample construction

Start: All individuals aged 25-60 in 2008. Drop individuals:

- … if not observed or if tenure is missing in *any* wave.
- … already in SH in Wave 1 (want observed moves).
- … who move multiple times (for simplicity).
- … who move after wave 9 (avoid “forbidden comparisons”).
- … who never move into SH.

Treatment: Move into SH in waves 1-5.

Control: Not yet moved + Move into SH in waves 6-9.

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Treatment – mention this?

How many waves to use?

Trade-off: Sample size vs estimating long-term effects.

- With T waves, we estimate T-1 pre-periods, T-2 post-periods.
- Only use individuals present in all T waves, but N falling due to attrition.

Period: Use *T* waves, balancing sample size with estimating long-term outcomes.

Treatment: Moving into SH, *conditional on being likely to move into SH*.

- Treatment group: Moving into SH in waves 1-*T*.
- Control group: Moving into SH in waves *T*-13

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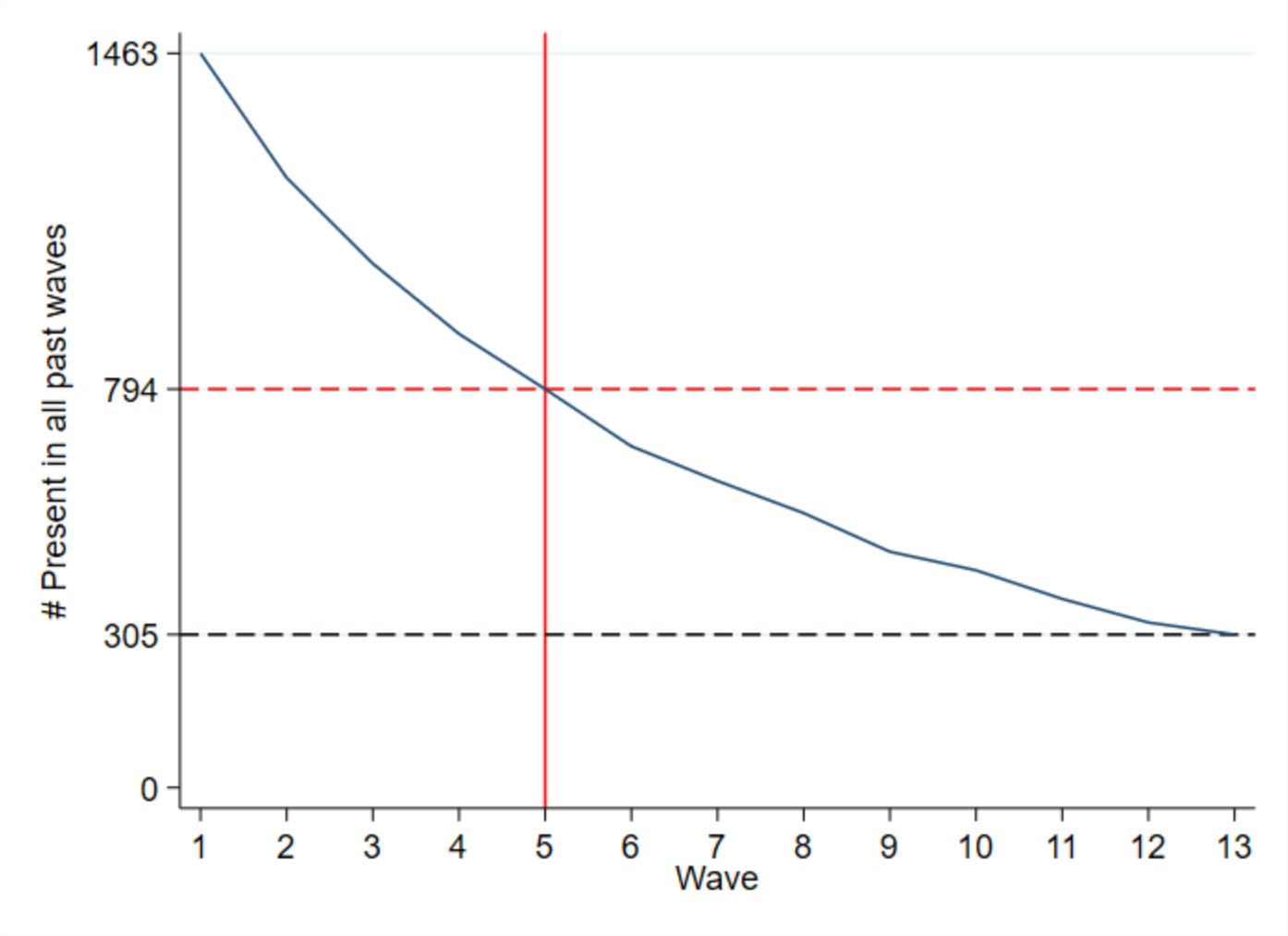
Potential Sample Size



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Extra Results

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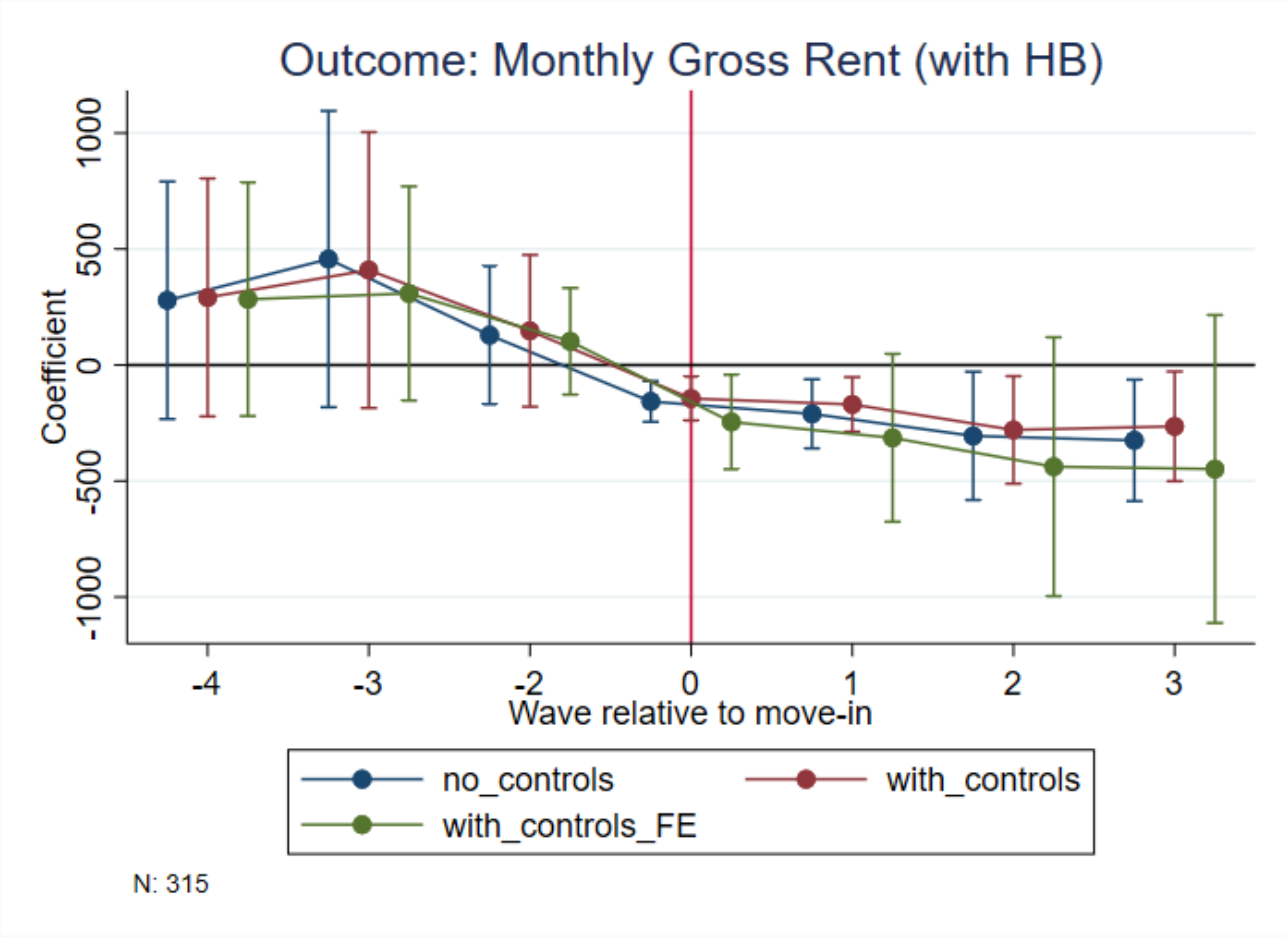
Gross Rent



Higher score = higher estimated chance of mental distress. A score of 3 is a commonly used threshold level.

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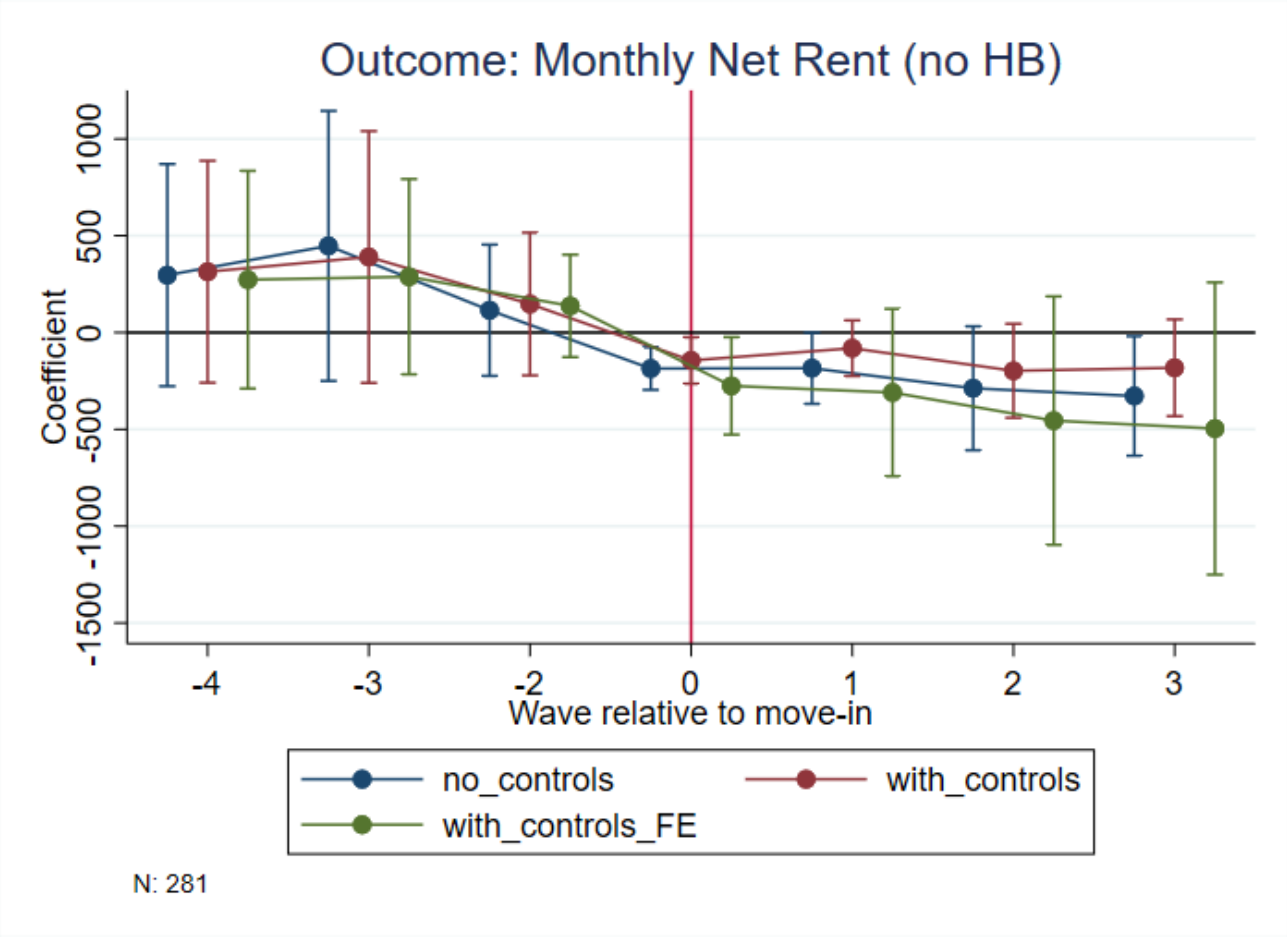
Net Rent



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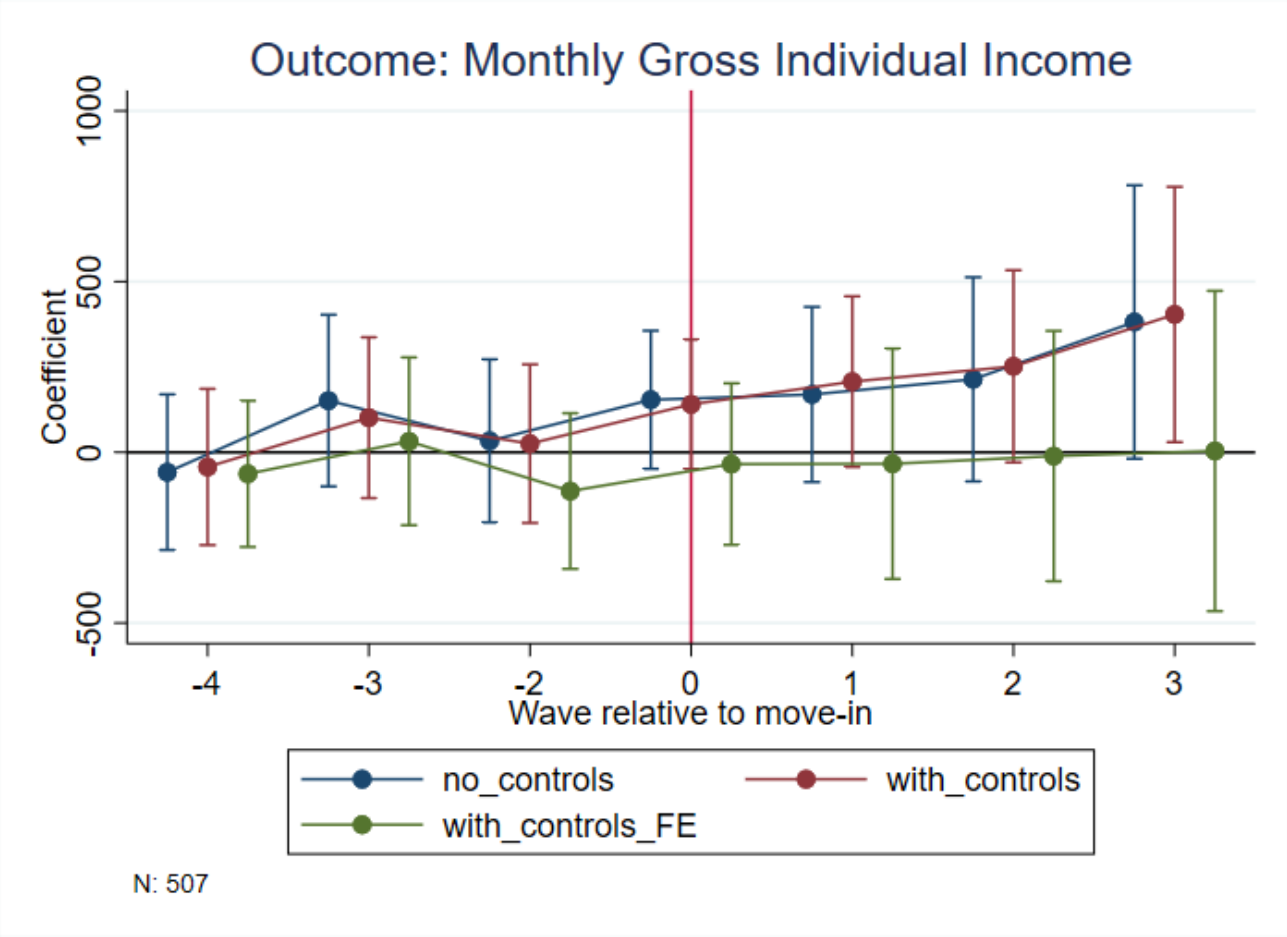
Monthly Individual Income



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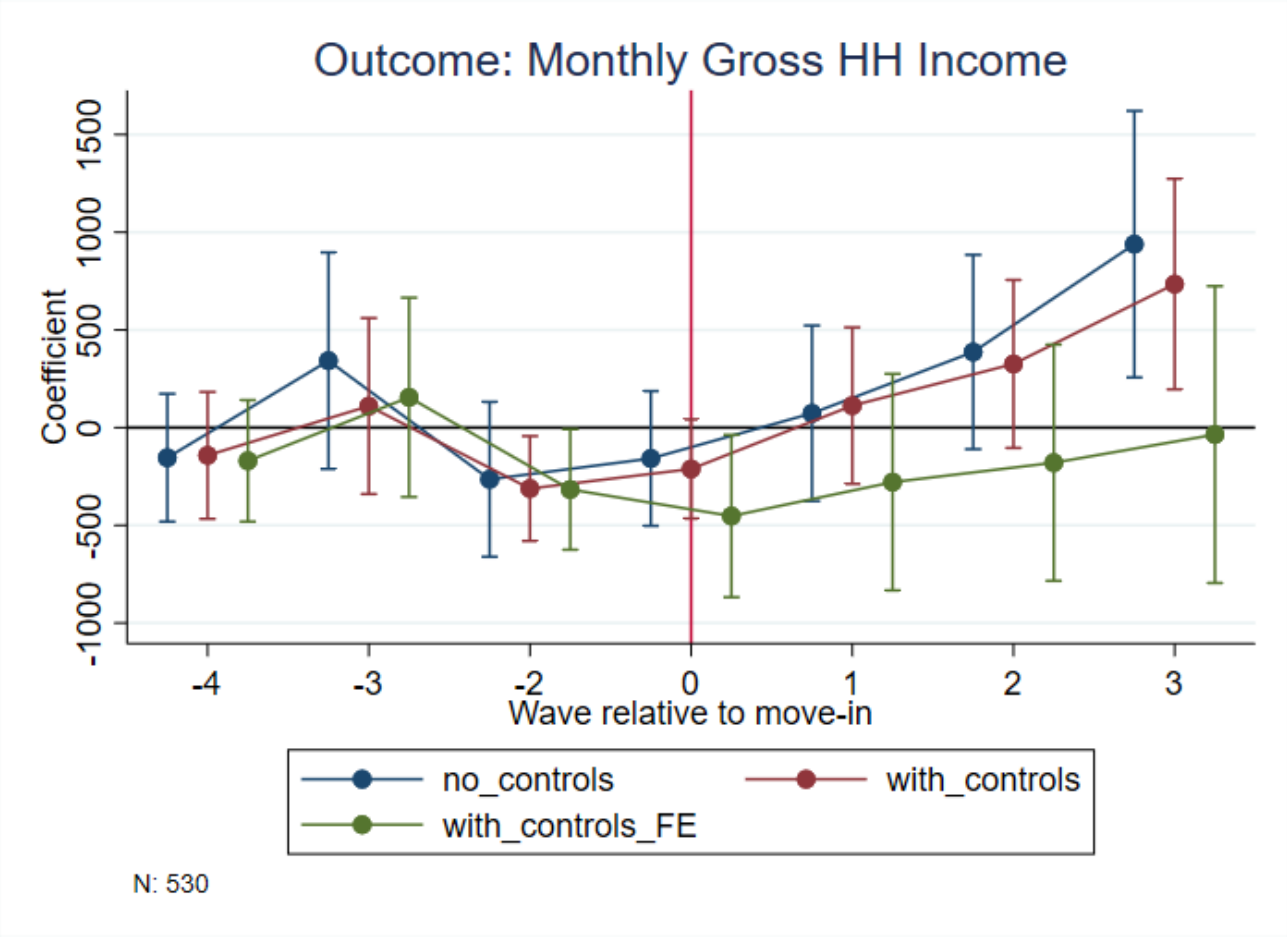
Household Income



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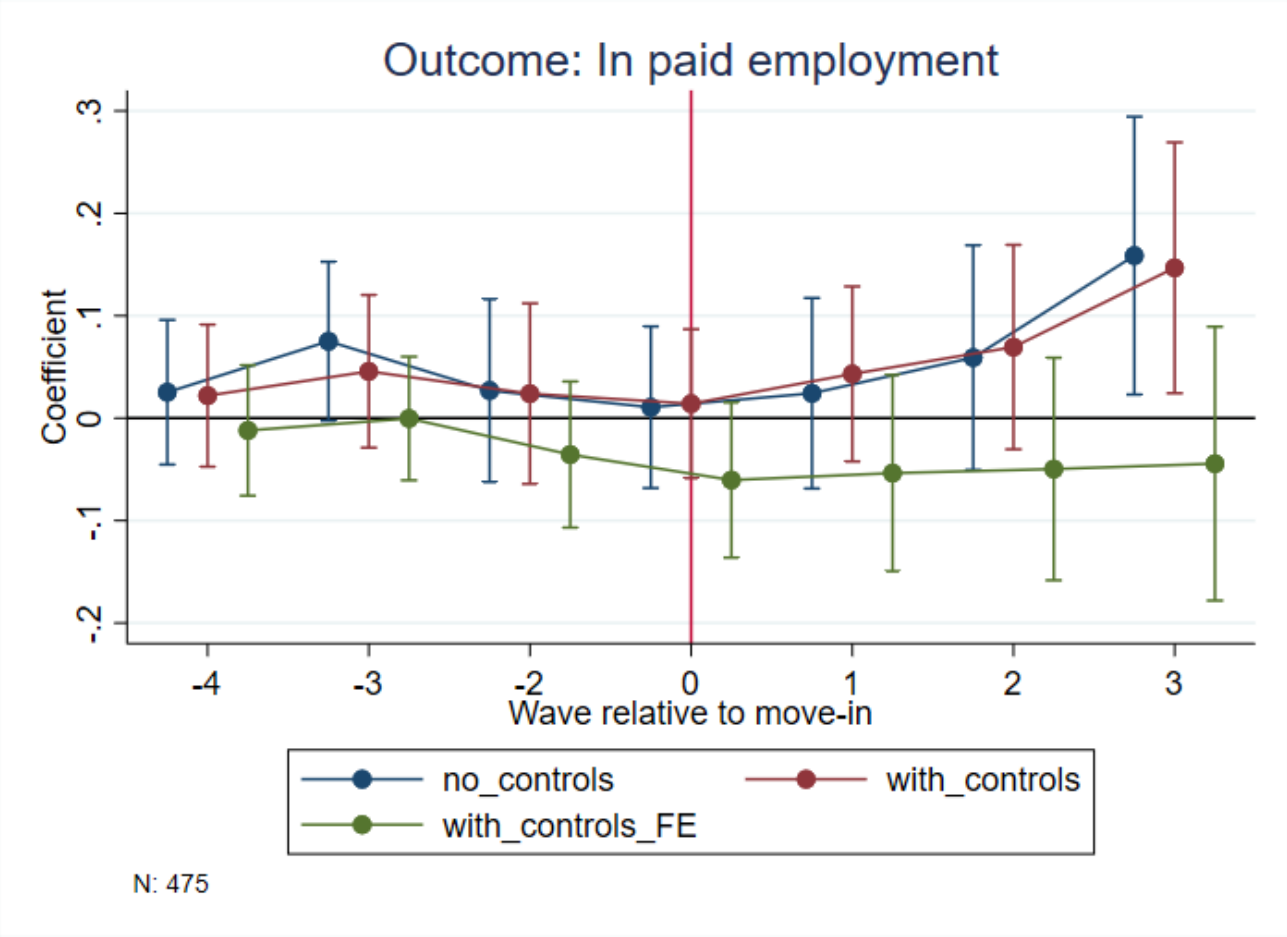
Employment



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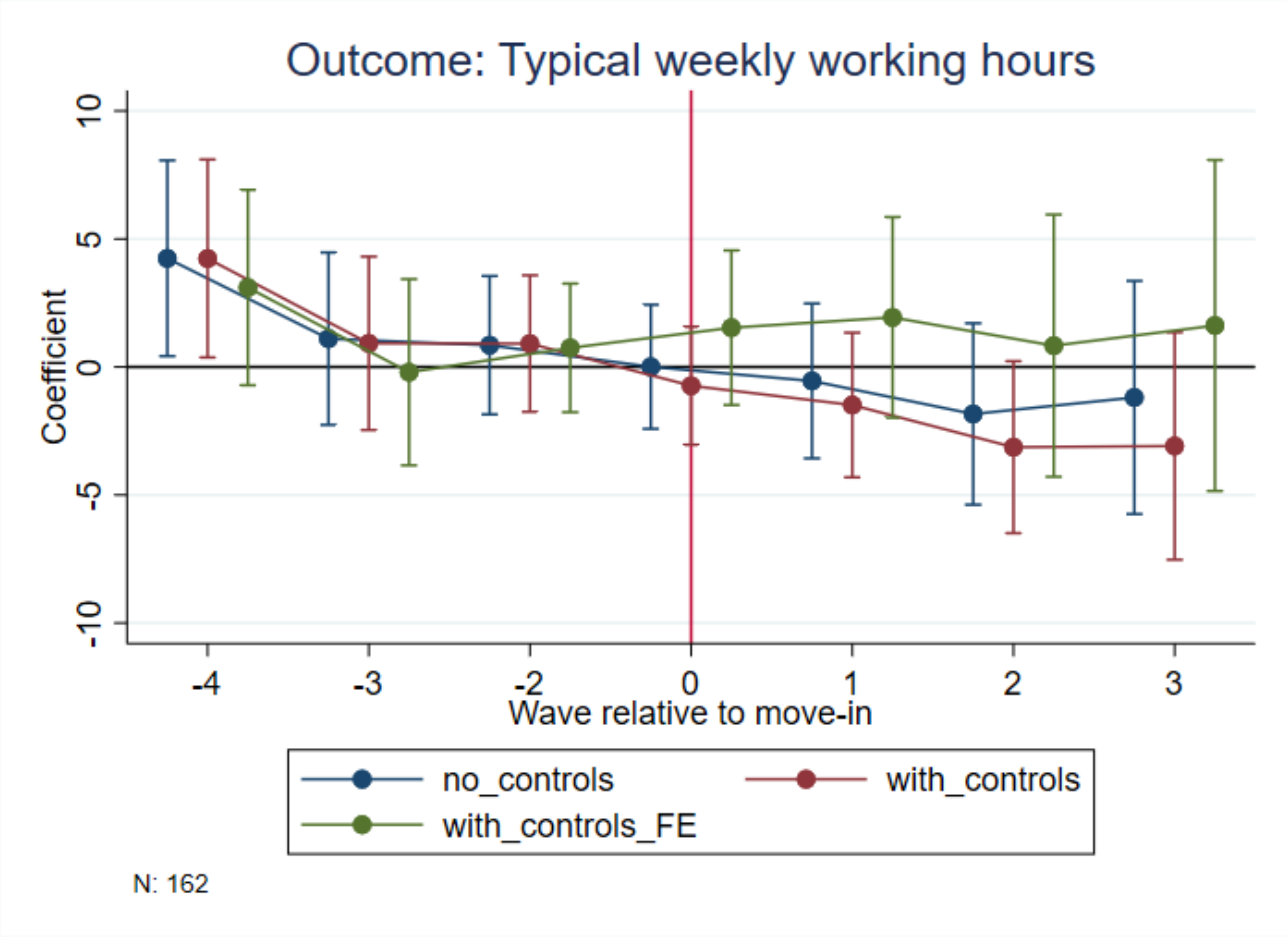

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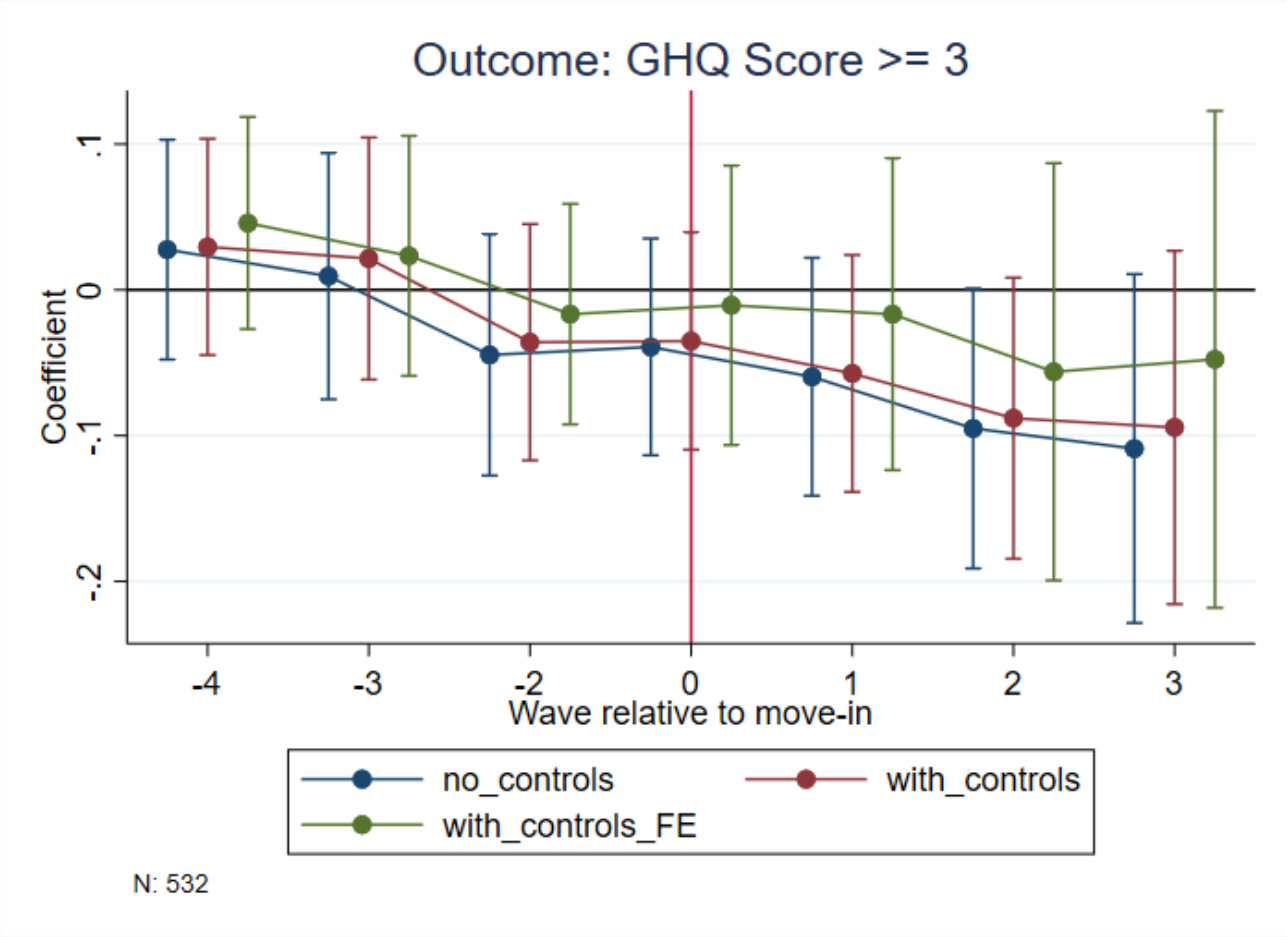

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